

assignment3 - distribution of charges electric field

 This is a preview of the published version of the quiz

Started: Mar 15 at 3:10pm

Quiz Instructions

It is EXTREMELY IMPORTANT, CRUCIAL that you HIDE the hints first and try hard. Use the hints as a last resort or to check your answers. There is a tutorial video in the shared dropbox.

IT is EXTREMELY important to open the textbook and study solved problems

review electric fields by a charge distribution (Calculus) . USe your textbook or:

[https://phys.libretexts.org/Bookshelves/University_Physics/Book%3A_Introductory_Physics_-_Building_Models_to_Describe_Our_World_\(Martin_Neary_Rinaldo_and_Woodman\)/16%3A_Electric_Charges_and_Fields/16.03%3A_The_Electric_Field](https://phys.libretexts.org/Bookshelves/University_Physics/Book%3A_Introductory_Physics_-_Building_Models_to_Describe_Our_World_(Martin_Neary_Rinaldo_and_Woodman)/16%3A_Electric_Charges_and_Fields/16.03%3A_The_Electric_Field)

<https://openstax.org/books/university-physics-volume-2/pages/5-5-calculating-electric-fields-of-charge-distributions>



Question 1 11.2 pts

Two parallel square metal plates that are 1.5 cm apart and 22 cm on each side carry equal but opposite charges uniformly spread out over their facing surfaces. How many excess electrons are on the negative surface if the electric field between the plates has a magnitude of 18,000 N/C? ($k = 1/4\pi\epsilon_0 = 9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$, $e = 1.6 \times 10^{-19} \text{ C}$)

hint: See my slides and lecture. $E = \sigma/\epsilon_0$. σ is the charge density Q/area . First find sigma (using equation $E = \sigma/\epsilon_0$). Then find Q using $Q = \sigma A$. with A is the area of the square (0.22×0.22). Then remember charges are quantized so $Q = Ne$ with e the charge of an electron.

☐ 7.7 10^{-9}

☐ 4.8 10^{10}

☐ 2 10^9

☐ 1.6 10^{-11}



Question 2 11.2 pts

A charge of 25 nC is uniformly distributed along a semi circle (radius = 2.0 m) .

What is the magnitude of the electric field at the center of the circle along which the arc lies?

hint: how nice. There is a video-tutorial in my shared dropbox (see assignment3)

but try first ! Check the methodology in my slides. please

☐ 81N/C

☐ 71N/C

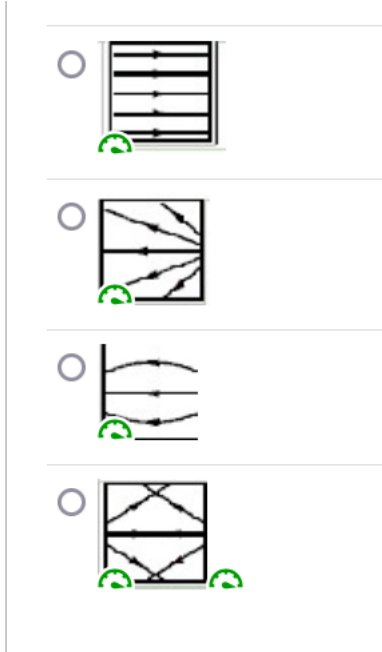
☐ 36N/C

☐ 26N/C



Question 3 11.2 pts

Which one of the diagrams below is *not* a possible electric field configuration for a region of space which does *not* contain any charges? A, B, C, D



hint: See rules slide 55

- ☐ C
- ☐ A
- ☐ D
- ☐ B
- ☐

Question 4 11.2 pts

A charged wire is placed along an x-axis from $x=0$ to $x=3$ m. The charge is distributed uniformly along the wire. The charge density is 9 nC/m . (so $1\text{ m} \rightarrow 9 \times 10^{-9}\text{ C}$). What is the electric field at a point P ($x=4$ m) ?

hint: First make a drawing. See textbook or examples in my slides. Use your calculus skills. Note that $1/(4\pi\epsilon_0)$ can be replaced by $K=9 \times 10^9$. Its easier. And $Q = \text{total charge} = \text{density} \times \text{length of the wire}$.

- ☐ 20 N/C
- ☐ 81 N/C
- ☐ 74 N/C
- ☐ 88 N/C
- ☐ 61 N/C
- ☐

Question 5 11.2 pts

You have a charged wire along the x-axis from 0 to 3 m. The charge density is 2 nC/m . You want the y-component of the electric field at the point P(0,4). Which of the following integral will give you the y-component of the electric field?

- a. $\int_0^3 \frac{72dx}{(16+x^2)^{3/2}}$
- b. $\int_0^3 \frac{18dx}{(16+x^2)^{3/2}}$
- c. $\int_0^3 \frac{72dx}{16+x^2}$
- d. $\int_3^0 \frac{18dx}{16+x^2}$
- e. none of these

Hint: Use your calculus skills. You have examples in my slides and the textbook.

$E_y(\text{net}) = \sum (dE_y)$. dE_y is from a small dq over a dx along the wire.

$dE_y = dE \sin(\alpha)$ with $\sin(\alpha) = y/d$ (figure out $d = \sqrt{x^2 + y^2}$.)

$dE = Kdq/(d^2)$ in magnitude. $dq = (\text{density}) \cdot dx$. The sum becomes an integral.

☐

b

☐

c

☐

d

☐

a

☐

none of these

⋮

Question 6 11.2 pts

Building on the previous question. Use wolfram alpha to find the integral

<https://www.wolframalpha.com/>  [\(https://www.wolframalpha.com/\)](https://www.wolframalpha.com/)

integral of for x between 0 and 3

☐

11.5N/C

☐

2.7 N/C

☐

0.675N/C

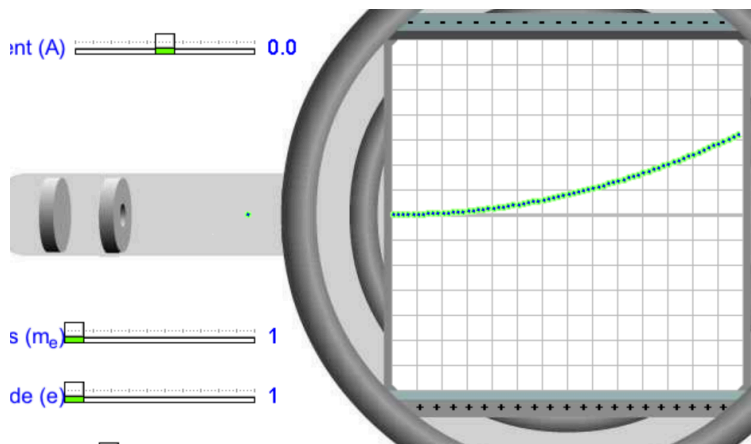
☐

2.9N/C

⋮

Question 7 11.2 pts

Consider our old friend the CRT. (like shown in class). A positive charge (4 mC) has a velocity along the x -axis (25m/s) and enters a uniform electric field (60N/C) pointing to the positive y -axis. The mass of the particle is 50 g . The charge is deflected. see below.



So E is up. The initial velocity is along the x -axis. What is the speed of the particle after 5s?

hint: convert g to kg and mC to C

Along the x -axis the velocity stays the same.

Along the y -axis there is an acceleration = force/mass = $qE/mass$

$$V_y = at$$

V_x is the same.

To find the speed use Pythagorean theorem. You have the x component and the y component

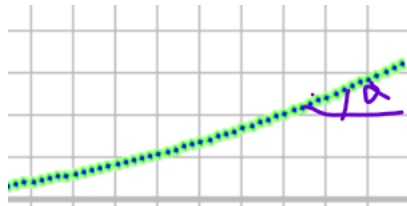
- ☐ 35m/s
- ☐ 44m/s
- ☐ 49m/s
- ☐ 24m/s
- ☐ 32m/s



Question 8 11.2 pts

Building on the previous question. You have the V_x and the V_y of the velocity.

What is the angle of deflection? about



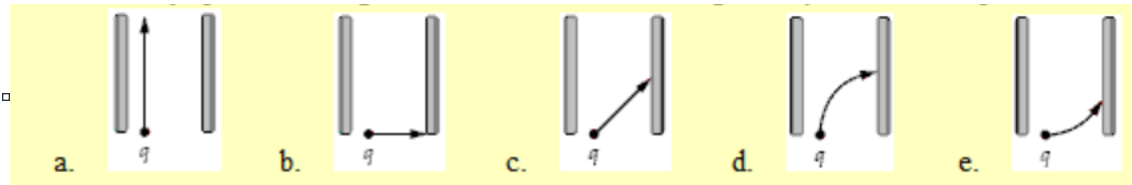
hint: remember vectors $\tan(\alpha) = V_y/V_x$

- ☐ 14 degrees
- ☐ 64 degrees
- ☐ 44 degrees
- ☐ 24 degrees



Question 9 11.2 pts

A positively charged particle is moving in the +y-direction when it enters a region with a uniform electric field pointing in the +x-direction. Which of the diagrams below shows its path while it is in the region where the electric field exists. The region with the field is the region between the plates bounding each figure. The field lines always point to the right. The x-direction is to the right; the y-direction is up.



- ☐ a
- ☐ e
- ☐ d
- ☐ c
- ☐ b

Not saved

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